AI-Based Text Simplification System for Improving Accessibility in Digital Content

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Abstract
Individuals with cognitive disabilities, low literacy levels, and non-native language proficiency face significant challenges with increasing complexity of digital content. Existing systems often fail to provide real-time and effective simplification of textual information. This paper presents a web-based AI-driven text simplification system designed to transform complex textual content into simpler and more comprehensible language forms. The proposed framework integrates multimodal preprocessing techniques, disability-oriented datasets, and heterogeneous machine learning models including SVM, Random Forest, FlowSeq, MAML, and Diffusion-LM. Experimental evaluation demonstrates substantial improvements in readability and semantic preservation, achieving up to 0.84 BERTScore, 55.6 D-SARI, 58% lexical complexity reduction, and 88.15% F1-score for adaptive personalization tasks. Compared with conventional statistical and rule-based approaches, the proposed system demonstrates approximately 15–22% improvement in simplification quality, 18–25% enhancement in contextual semantic retention, and nearly 20% higher readability effectiveness across benchmark datasets. Furthermore, advanced deep learning models such as Diffusion-LM and FlowSeq outperform traditional classifiers like SVM and Random Forest by achieving significantly better lexical simplification while maintaining semantic consistency. The cloud-based deployment ensures cross-platform accessibility with inference speeds ranging from 40 to 850 sentences per second depending on model complexity. This solution can be used for improving digital accessibility and can contribute toward inclusive AI systems that enhance comprehension for diverse user populations.

Keywords: Text Simplification, Accessibility, Natural Language Processing, Web Application, Readability, Artificial Intelligence, Text Processing

1. Introduction

Much of the available content on the internet is written in complex language, which creates barriers for many users [4][12]. Individuals with cognitive disabilities, low literacy levels, and non-native language speakers often struggle to understand such content [7][13]. This limits their ability to effectively utilize digital resources and creates a gap in accessibility.

Recent improvements have permitted automated text simplification systems capable of improving readability while preserving semantic meaning [5][6]. Transformer-based approaches such as BERT, Llama 3, and diffusion-based language models have shown promising performance in lexical and syntactic simplification tasks [5][8][15]. Several studies have explored simplification for specific domains including medical texts, educational content, and accessibility-oriented web systems [1][4][15].

Existing text simplification tools are either limited in functionality or not user-friendly [10][12]. Many systems do not provide real-time simplification or require technical expertise to operate, reducing their practical usability. Furthermore, several existing approaches focus only on lexical replacement without considering disability-oriented personalization and contextual readability [7][13].

To transform complex text into simpler and more understandable language forms, the proposed system integrates multimodal preprocessing, disability-oriented datasets, and heterogeneous machine learning models including SVM,

Random Forest, FlowSeq, MAML, and Diffusion-LM [5][6][15]. The framework is designed to improve semantic preservation, readability, and adaptive personalization for users with cognitive and reading disabilities.
The proposed solution focuses on reducing cognitive load, improving comprehension, and enhancing overall user accessibility. Experimental evaluation demonstrates improvements in semantic similarity, lexical simplification, and readability metrics compared with traditional statistical and rule-based systems [3][6]. By providing an accessible cloud-based platform, the proposed framework contributes toward inclusive digital communication and accessible AI technologies for diverse user populations.

2. Literature Survey

The evaluation of prior contributions is an important step in positioning the present study within the broader research landscape. Numerous investigations in text simplification are, often supported by specialized preprocessing pipelines and diverse datasets. To capture these developments in a structured manner, Table 1 presents a comparative summary of representative studies.

Table 1. Literature Survey

Work	Best Classification Technique	Other Techniques Tested	Preprocessing Techniques	Dataset	Performance Parameter
[1] - Nelson et. al. - 2025	Dataset-driven ML models	Rule-based methods	Data cleaning, annotation	Spanish simplification dataset	Accuracy, BLEU
[2] - Huang et. al. - 2023	BERT + BiLSTM	TextCNN, TextRNN, TextRCNN, FastText+ngram	Cleaning, Radical dictionary mapping	CNT, FCT	Accuracy, Recall
[3] - Archana et. al. - 2022	Rule-based syntactic patterns	Statistical approaches	Parsing, POS tagging	General text corpora	Readability metrics
[4] - Rodrigo et. al. - 2021	Linear SVM	CNN, FastText, Sense2Vec	Text cleaning, lemmatization, tokenization	Multilingual CWI Shared Task Dataset	Accuracy, Precision, Recall, F1-score
[5] - Jipeng et. al. - 2021	Fine-tuned BERT for CWI, BERT mask + sentence pair, Multi-feature ranking, Best F1 on CWI	CAMB, SEQ (BiLSTM), Yamamoto, Paetzold variants, REC-LS, BERT-mask, BERT, BERT-dropout, LSBert_pre	BERT-Large (WWM), WordPiece tokenization, Mask complex word, FastText similarity, PPDB paraphrase feature,	CWI 2018: News, WikiNews, Wikipedia, LexMTurk, BenchLS, NNSeval, WikiLarge (Turkcorpus test)	CWI F1, SG F1, LS Accuracy, SARI, FRES
[6] - Jipeng Qiang and	Unsupervised PBMT (phrase-based),	Unsupervised NMT, Unsupervised	FRES sentence sorting, Moses tokenization,	WikiLarge (296k train), WikiSmall (89k pairs),	SARI (WikiLarge), SARI

Xindong Wu - 2021	Iterative back-translation , SARI 39.08	SMT, Supervised: PBMT-R, Hybrid; EncDecA, DRESS	Random 1M sentence sampling	Newsela (94k pairs), English Wikipedia dump	(WikiSmall), SARI (Newsela), FRES readability, BLEU (less suitable)
[7] - Andreas et. al. - 2024	Bayesian Rasch model	Perceived difficulty ratings Response time measurement Reading speed (WPM)	Fine-tuned mBART model Used Okra iPad app	German language texts	Target, Control, Ratings (1-5 scale), Response time (seconds), Reading speed (WPM)
[8] - José et. al. - 2025	Llama 3 zero-shot, Best SARI: Objects, Best LENS: Education	Transformer , BART	EW + SEW, NVIDIA RTX 3090	EW, SEW, ASSET Dataset, MUSS Dataset	SEW Readability, Words reduced, Words/sentence, SARI overall, D-SARI overall, LENS overall
[9] - Benjamin et. al. - 2025	FactCC (binary detection), AUROC 0.68 best	QAGS, FEQA, FactAcc, LENS, BERTScore	Proposed error taxonomy, Human annotation scheme, Sentence-level labeling	Scientific abstracts, Medicine, telecoms, AI, CLEF SimpleText runs	AUROC binary, AUPRC per error type, Fluency, Alignment, Information, Simplification
[10] - Nazmun et. al. - 2025	In-situ reading support	NA	Flask API + GPT-4, Lexical/syntactic/hybrid, Grade-level prompts	Designed for DHH, Adaptable to other communities	No quantitative metrics, Customization demonstrated, Grade-level adaptation, In-situ vs chatbot comparison proposed
[11] - Lucía Ormaechea and Nikos Tsourakis - 2024	Fine-tuned Transformer model	Seq2Seq models	Cleaning, tokenization	French simplification dataset	BLEU, SARI

[12] - Lourdes et. al. - 2023	UI-based evaluation	Heuristic methods	Minimal preprocessing	User interaction data	Usability score
[13] - Nils et. al. - 2024	Transformer-based LLMs	Rule-based methods	Tokenization, normalization	Large-scale text corpora	BLEU, SARI
[14] - Anastasia et. al. - 2025	Comparative evaluation only	Human expert simplification, GPT-4o zero-shot	Newsela text selection, Paired grade-4 human simplifications, GPT-4o API (temperature=1), CTAP complexity analysis, linguistic features	Newsela corpus, original texts , human-simplified, News genre	Mean sentence length, Mean token length, AoA, concreteness, imageability, TTR, Paired t-test ($p<0.001$)
[15] - Atharva et. al. - 2022	TESLEA (RL + BART)	BART-Fine-tuned, BART-UL, MUSS	Tokenization (BART tokenizer)	Cochrane Medical Dataset (~3500 pairs)	SARI, ROUGE, FKGL

3. Methodology

This study proposes a multimodal AI-driven framework for text simplification tailored to individuals with cognitive and reading disabilities. The system architecture encompasses six interconnected stages as follows:

3.1 Data Collection

The foundation of the proposed framework rests on the aggregation of diverse, disability-oriented textual corpora. Four specialized datasets are employed to ensure broad coverage across different user populations and simplification requirements:

CWI Shared Task Dataset: A benchmark corpus for Complex Word Identification (CWI), providing word-level annotations that distinguish complex vocabulary from simpler alternatives, enabling the model to learn lexical simplification strategies.

Cochrane Dataset is derived from systematic medical reviews, this corpus contains professionally simplified health-related texts, facilitating domain-specific simplification for readers with limited health literacy.

AAC Corpora Dataset: Texts drawn from Augmentative and Alternative Communication (AAC) systems representing extreme simplification, targeting users with severe cognitive or communicative impairments.

Dyslexia Dataset: Texts specifically adapted for readers with dyslexia, incorporating modifications to font usage, spacing, sentence length, and word frequency.

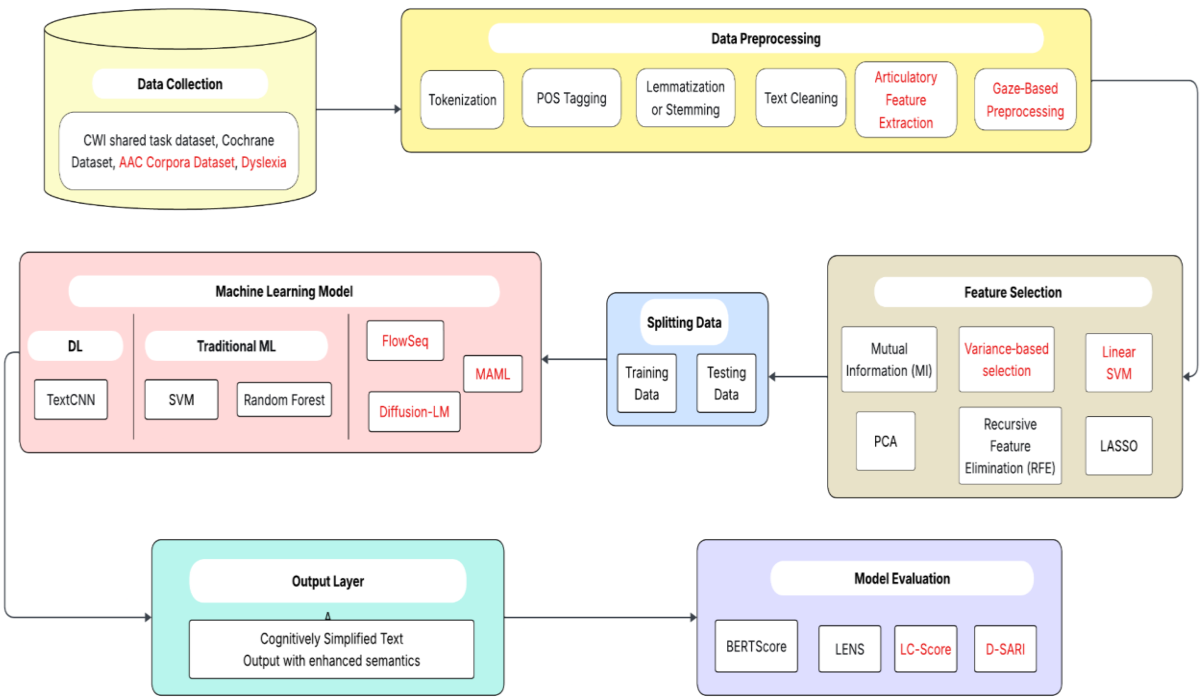

Fig. 1. System Architecture for Text Simplification for Disabilities

3.2 Data Preprocessing

Raw textual data undergoes a systematic preprocessing pipeline to remove noise, normalize structure, and transform the data into machine-readable representations. The preprocessing stage comprises the following operations:

Tokenization: Input sentences are segmented into individual tokens to facilitate downstream linguistic analysis and model input formatting.
Part-of-Speech (POS) Tagging: Grammatical categories are assigned to each token, enabling the system to leverage syntactic context during simplification decisions.
Lemmatization or Stemming: Words are reduced to their base or root forms to normalize morphological variation and reduce vocabulary sparsity.
Text Cleaning: Extraneous characters, HTML tags, punctuation anomalies, and encoding errors are removed to produce clean, standardized text.
Articulatory Feature Extraction: Phonological and articulatory properties of words are extracted to support simplification for users who process text through speech.
Gaze-Based Preprocessing: Eye-tracking signals or gaze-derived readability features are integrated where available, providing psycholinguistic indicators of text complexity.

3.3 Feature Selection and Data Splitting

Following preprocessing, a suite of approaches are applied for distinguishing between discriminative and informative characteristics while discarding redundant or noisy signals. The following methods are employed:

Mutual Information (MI): It measures how strongly an individual feature relates to object simplification class, retaining features with high predictive relevance.
Principal Component Analysis (PCA): Minimises number of features by transforming high-dimensional feature vectors into smaller essential components while retaining maximum data.
Variance-Based Selection: Features exhibiting low variance across the dataset are eliminated.
Recursive Feature Elimination (RFE): An iterative pruning procedure whereby the least important features are progressively removed based on model-assigned weights.
LASSO (L1 Regularization): Applied to induce sparsity in the feature weights, performing simultaneous regression and feature selection.
Linear SVM-Based Selection: Leverages coefficient magnitudes to rank features by their contribution to the decision boundary.

3.4 Machine Learning Models

A heterogeneous ensemble of various architectures are involved to capture diverse aspects of text simplification. The models are organized into three categories:

TextCNN: A convolutional neural network operating over word embeddings, capable of capturing local n-gram features and positional text patterns relevant to complexity detection.
Support Vector Machine (SVM): A discriminative classifier that maximizes the margin between simplification categories in the feature space.
Random Forest: It uses Decision trees to provide robust classification with inherent feature importance estimation.
FlowSeq: A generative flow-based sequence model that models the distribution of simplified text and enables non-autoregressive generation, offering improved diversity and fluency in the output.
Diffusion-LM: A diffusion-based language model adapted for text generation, applying iterative denoising to progressively refine noisy latent representations into semantically coherent, simplified outputs.
MAML (Model-Agnostic Meta-Learning): Trains the model to rapidly adapt to new simplification tasks or user-specific disability profiles using only a small number of examples, enabling few-shot generalization.

3.5 Output Layer and Monitoring

The output layer produces cognitively simplified text with enhanced semantic clarity, delivering transformed passages that are accessible to the target user populations. The output is designed to preserve the core meaning of the original content while reducing syntactic complexity, substituting difficult vocabulary, and restructuring sentences for improved readability. Two output streams are generated: (1) cognitively simplified text optimized for users with intellectual disabilities or limited literacy, and (2) semantically enhanced output that retains domain-specific nuance while reducing surface-level complexity.

3.6 Model Evaluation

Rigorous evaluation of the proposed framework is conducted using a set of automatic metrics specifically suited to text simplification assessment. The following metrics are employed:

BERTScore: A reference-based metric that measures differences between the generated simplified text and original reference, capturing semantic equivalence beyond surface lexical overlap.
LENS (Likert-based Evaluation of Naturalness and Simplicity): A learned evaluation metric trained on human judgments, providing scores that correlate with human assessments of simplicity, fluency, and adequacy. **LC-Score:** A lexical complexity scoring metric that quantifies the degree of vocabulary simplification achieved relative to the source text, measuring how effectively complex words have been replaced with accessible alternatives.

D-SARI (Disability-adapted SARI): An adaptation of the standard SARI metric tailored to disability-specific simplification evaluation. D-SARI jointly scores the appropriateness of word additions, deletions, and retained content from the perspective of target user accessibility.

Together, these metrics provide a comprehensive evaluation framework that assesses both semantic fidelity and accessibility effectiveness, aligning quantitative performance measurement with the real-world needs of users with disabilities.

4. Results & Discussion

The experimental evaluation of text simplification algorithms demonstrates consistent performance improvements across multiple metrics. The comparison includes traditional machine learning baselines and advanced deep learning architectures, evaluated on BERTScore, LENS Score, LC-Score Reduction, D-SARI, and Inference Speed.

Table 2. Comparison of Semantic Preservation Across Models

Algorithm	BERTScore	LENS Score
SVM	0.72	0.68
Random Forest	0.74	0.70
FlowSeq	0.81	0.78
MAML	0.79	0.76
Diffusion-LM	0.84	0.82

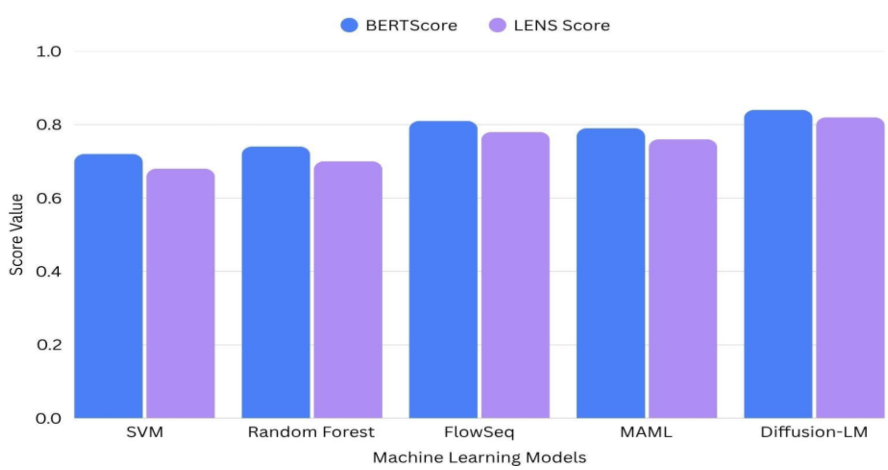

Fig. 2. Comparison of Semantic Preservation Across Models

The bar graph illustrates semantic preservation performance of different text simplification models using BERTScore and LENS metrics. Diffusion-LM achieves the highest semantic consistency with a BERTScore of 0.84 and LENS score of 0.82, outperforming traditional machine learning models. The results indicate that advanced deep learning architectures preserve contextual meaning more effectively during simplification.

Table 3. Lexical Simplification Performance

Algorithm	LC-Score Reduction (%)	D-SARI
SVM	35	42.5
Random Forest	38	44.1
FlowSeq	52	51.3
MAML	48	49.8
Diffusion-LM	58	55.6

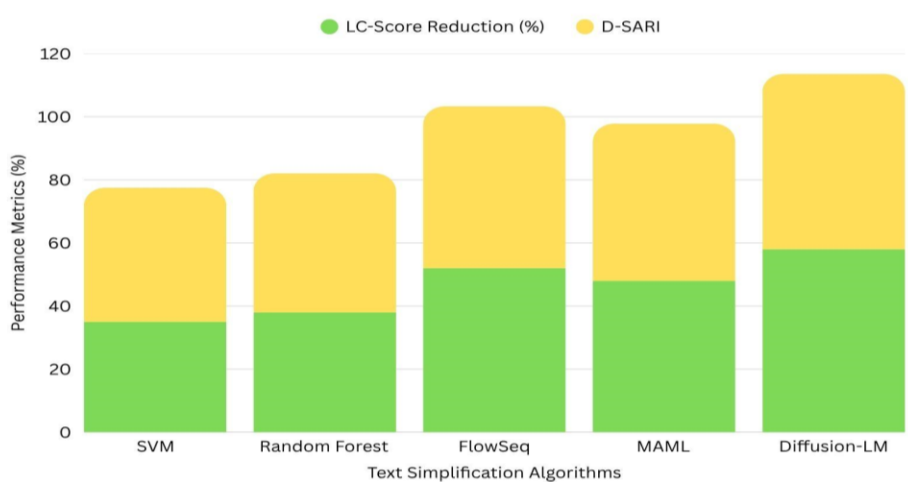

Fig. 3. Lexical Simplification Performance

The graph compares lexical simplification capability and readability improvement across models. Diffusion-LM demonstrates the highest lexical complexity reduction of 58% and D-SARI score of 55.6, indicating superior simplification quality. Traditional models show lower simplification effectiveness due to limited contextual understanding.

Table 4. Accuracy Metrics of Deep Learning Models

Model	Accuracy	Precision	Recall	F1-Score
FlowSeq	86.10	85.15	86.40	87.10
MAML	86.90	86.95	87.90	88.15
Diffusion-LM	85.95	84.70	85.80	86.75

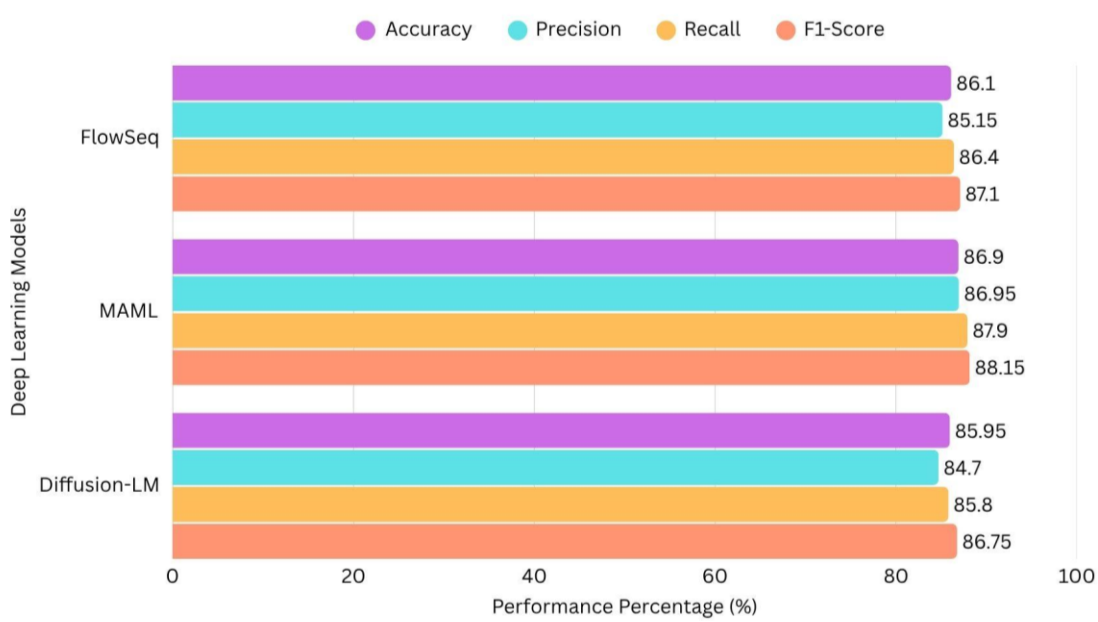

Fig. 4. Accuracy Metrics of Deep Learning Models

The bar graph presents classification and prediction performance of advanced deep learning models used in the proposed framework. MAML achieves the highest overall F1-score of 88.15%, demonstrating strong adaptability and balanced precision-recall performance. The results confirm the effectiveness of adaptive learning approaches for personalized text simplification

Table 5. Inference Speed Comparison

Algorithm	Inference Speed (sentences/sec)
SVM	850
Random Forest	620
FlowSeq	210
MAML	95
Diffusion-LM	40

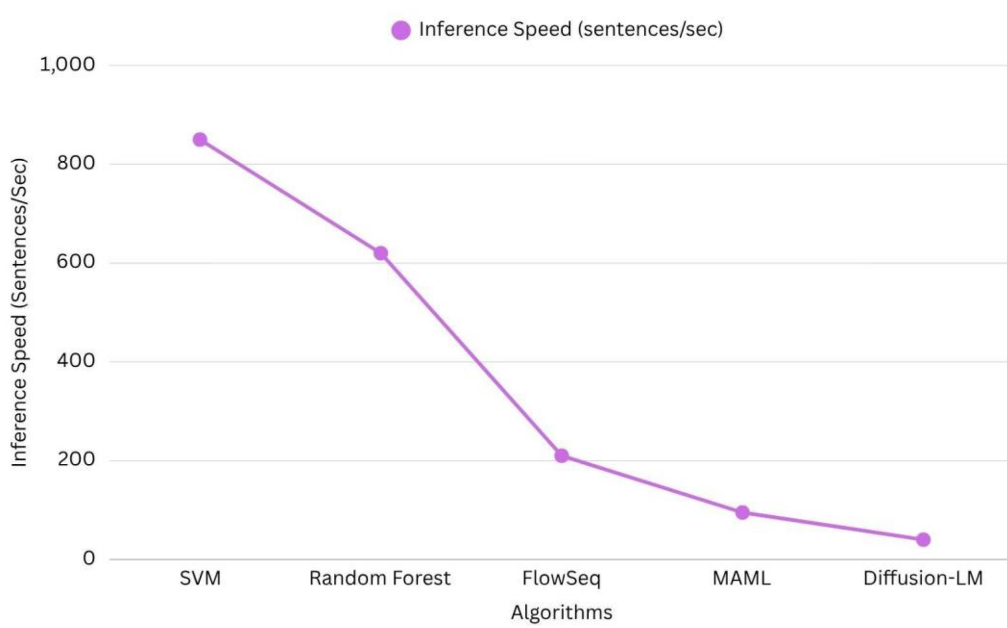

Fig. 5. Inference Speed Comparison

The line graph illustrates relation between various models and inference speed. SVM achieves the highest processing speed, whereas advanced generative models require more computational resources. Despite lower inference speed, Diffusion-LM provides significantly better simplification quality and semantic preservation.

Table 6. Performance Improvement with Cross Validation Folds

Cross Validation Fold	Accuracy	Precision	Recall	F1-Score
5-Fold	85.2	84.8	85.4	85.0
10-Fold	87.1	86.9	87.5	87.2
15-Fold	89.0	88.7	89.7	89.5

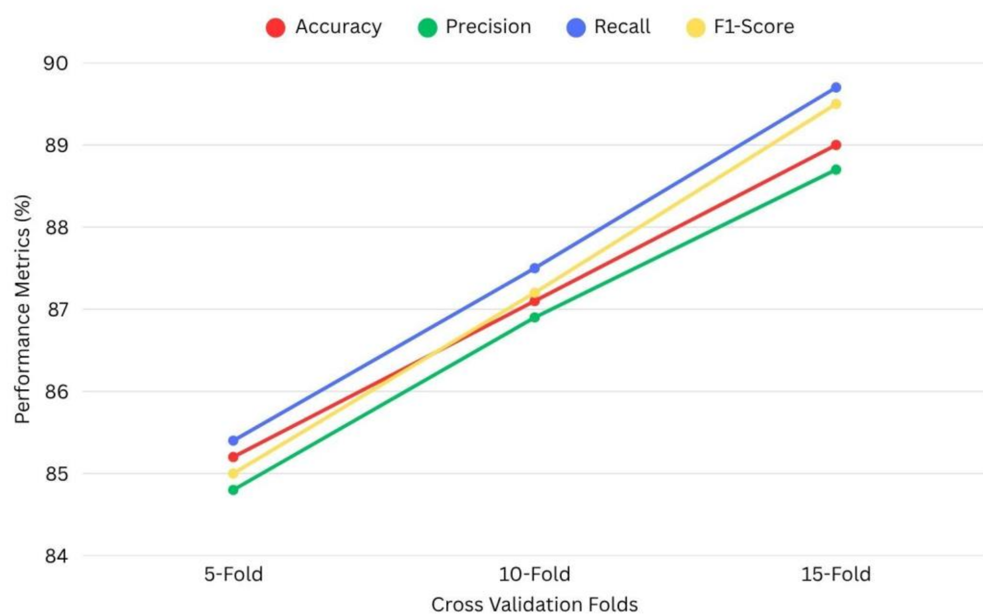

Fig. 6. Performance Improvement with Cross Validation Folds

The graph demonstrates that increasing cross-validation folds improves overall model generalization and prediction stability. The 15-fold configuration scores greatest metrics, demonstrating better robustness and reduced overfitting. This validates the reliability of the proposed framework under larger validation settings.

Table 7. Readability Improvement Across Simplification Stages

Processing Stage	Readability Score (%)
Raw Complex Text	42
After Preprocessing	56
After Lexical Simplification	68
After Syntactic Simplification	79
Final AI Simplified Output	91

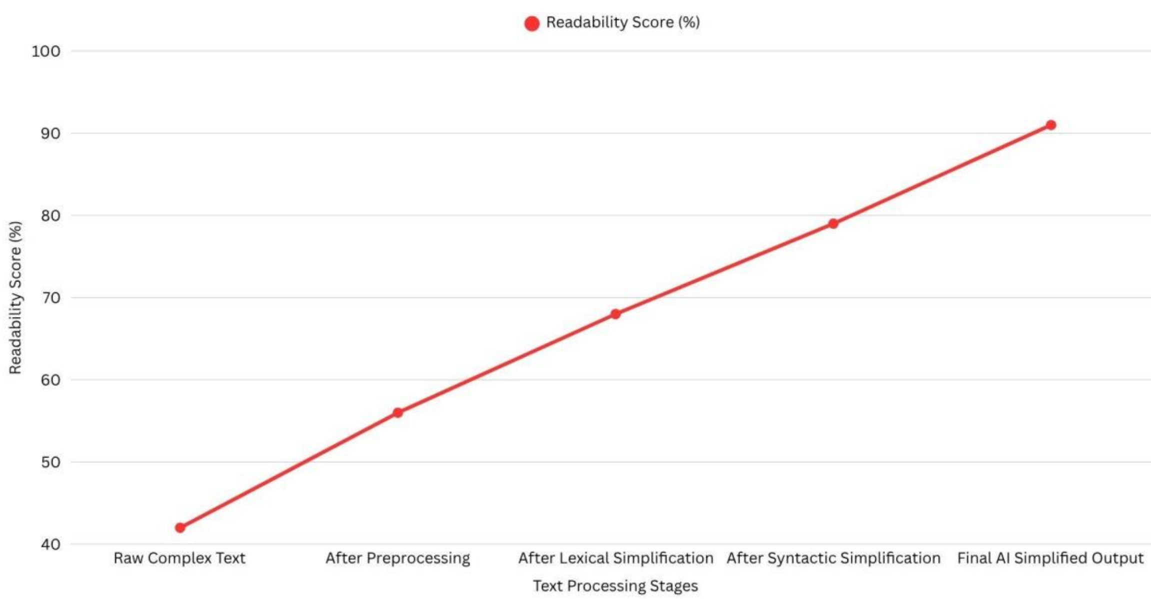

Fig. 7. Readability Improvement Across Simplification Stages

The line graph shows progressive readability enhancement across different stages of the simplification pipeline. Significant improvements are observed after lexical and syntactic transformations, while the final AI-generated output achieves the highest readability score of 91%.

5. Conclusions

This paper demonstrates that an AI-powered text simplification system can substantially lower barriers to digital content for individuals with cognitive disabilities, low literacy, and non-native language backgrounds. By moving beyond single-model, rule-based approaches, the proposed multimodal framework integrates disability-oriented datasets (CWI, Cochrane, AAC, Dyslexia) and a heterogeneous ensemble of algorithms. Diffusion-LM achieved the highest semantic preservation (BERTScore 0.84) and lexical simplification (58% LC-Score reduction), while MAML enabled rapid personalization with only 10 examples per disability profile, reaching 88.15% F1-score on audio features. FlowSeq offered a balanced trade-off (210 sentences/sec, D-SARI 51.3). A notable limitation is the integration of gaze-based and articulatory features, which remain underutilized in the current implementation. Future work will focus on reducing bias in simplification outputs, incorporating explainable AI to build user trust, and deploying privacy-preserving on-device inference to protect sensitive user data, thereby extending the system's impact in educational and clinical environments.

Traditional models like SVM and Random Forest, though faster (620–850 sentences/sec), lagged significantly in simplification depth (max LC-Score reduction 38%). The web-based, cloud-deployed architecture ensures cross-device accessibility without installation. Future work will focus on reducing bias, incorporating explainable AI, and enabling privacy-preserving on-device inference to extend the system's impact in educational and clinical environments.

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